

Factors Influencing Cloud-Computing Adoption in Small and Medium E-Commerce Enterprises in Jordan

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ABSTRACT

Adoption of cloud computing provides great benefits especially to e-commerce small and medium enterprises (SME) in the developing countries. as such businesses need to serve their customers and business partners effectively. This entails companies to use strong infrastructures that offer latest technologies in terms of hardware and software. However, the economical tough conditions may hurdle such endeavor and therefore SME need to adopt new solutions. Cloud computing as a new IT capability can solve many problems to SME in terms of cost, technology, and availability. Many SMEs in developing countries are still reluctant to adopt such technology. This research investigates the main factors that impact adoption of cloud computing by interviewing 16 managers of e-commerce SME and analyzes the collected data qualitatively. Results show that need, cost, security, and reliability are major factors that impact cloud computing adoption.

CCS Concepts

Social and professional topics→Professional topics→Computing and business→Socio-technical systems

Keywords

Cloud computing; E-commerce; need; security, adoption; IT architecture.

1. INTRODUCTION

Cloud computing is defined by NIST, 2009 as "a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction [1]. Cloud computing is being applied to many vital

disciplines as a new efficient technology, and e-commerce is an ideal one to be motivated by the features of cloud computing in saving the cost of IT infrastructure and respond quickly to the continuous changes in market demand, since it can be allowed to access the IT resources such as hardware, software, data and infrastructure on the cloud platform and pay for them as services. This facilitates the ability of e-commerce SMEs to enter the international markets easily and focus on their core business management, while the cloud service provider supply almost all the necessary products and services to the online store [2]. Small and medium-sized enterprises (SME) are identified in [32] as those companies who are employing less than 100 employees. In September 2005, the Prime Minister of Jordan endorsed the definition of industrial small enterprises as companies which employed between 10 and 49 employees and have a registered capital of JD 30,000 or more. Furthermore, medium companies employed from 50 to 249 employees and have a registered capital of JD 30,000 or more. SMEs accounted for 99.4 percent of Jordan's 156,000 enterprises, and account for 70 percent of the workforce, Cloud computing as a new trend in the IT domain can be considered as an important capability for businesses from two main pillars: IT efficiency and agility [29]. The two pillars are dominant factors for businesses to achieve competitive edge in competitive environments like e-commerce. Based on this, any company needs to gain the competitive edge; it should build a smart IT infrastructure that reflects its IT architecture [30]. The later should be carefully designed to align business needs without adding any new unnecessary technologies or devices. In response to this fact, e-commerce enterprises increasingly understand the importance of cloud computing and start adopting it in an enormous adoption rate.

E-commerce enterprises increasingly understand the importance of cloud computing and start adopting it in an enormous adoption rate. The technology adoption studies goal is to find and determine the main factors that affect organizations' decisions to accept and use such technology innovation [3]. Previous studies in this field show a lot of variations in the main factors that affect the adoption as well as the adoption rate among e-commerce SME [4]. Adopting cloud computing by SME in Jordan is not widely addressed by researchers and not yet completely understood. On the other hand there is a lack of theoretical models that investigates and analyzes the major factors that affect managers' decisions towards adoption of this technology in their organizations to meet current and future organizational needs.

Dynes [5] mentioned that IT managers are the dominant factor in organizational decision making regarding the adoption of new IT

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technologies like cloud computing. In response the main purpose of conducting this research is to explore and investigate the main factors that affect SMEs managers' decision towards adoption of cloud computing in their enterprises. The remaining sections of this article are structures as follows: section two presents the literature review. Section three discussed the theoretical background and the proposed model. Section 4 and 5 present the research methodology. Section 6 presents data analysis research results. Conclusions and future works are presented in section 7 followed by references.

2. LITERATURE REVIEW

Wang in [1] explained the adoption of cloud computing in e-commerce companies in view of three modes: service, business and technical. In the service mode e-commerce enterprises can access the required infrastructure which consists of needed hardware and software and pay for this service. So that data is saved remotely. This mode offers many advantages like decreasing the development costs, company operations, and maintenance. This yields to avoid companies dealing and owning a complex IT architecture. In addition, this helps companies to manage and run its daily tasks according to a given plan based on business priorities. xIn addition, there are many main forces for companies to involve cloud computing long-term strategies: some of these factors are related to requests to improve the efficiency of customer's services, decreasing costs, more flexibility is needed, , establishing of data centers, data integration and best processing to increase the efficiency of customer's online services. Furthermore offering customers with high quality of e-commerce products and services is another need that makes the e-commerce

enterprises adopt their cloud strategies [1]. Technically, when this strategy is adopted into an e-commerce company; one service provider can provide almost all the services to an e-commerce website. Hence, the supply chain of an e-commerce company will be changed [1].

Understanding advantages of cloud computing pinpoint the extent of adoption of cloud computing and identify the tasks or activities where it is possible to improve performances of the organizational tasks or functions. Generally, many previous researches have focused on cloud computing benefits and possible services. Studies [6], [7] pointed that cloud computing supports cost savings, scalability, availability, innovation speed, efficiency, easy management and disaster management. On the other hand, cloud computing adoption is faced with a number of challenges and difficulties; these challenges starting with trust between clients and vendors, because cloud computing calls for organizations to trust vendors with the management of their IT resources and data [8]. In addition, adopting of cloud computing might be faced with some structural aspects that include complex structure and compatibility [9]. Table 1 below lists some previous studies with factors that are used by managers in the adoption of cloud computing. Saleh in study [10] proposed a framework for adopting of cloud computing in e-commerce enterprises. His suggested framework highlights the important role of cloud computing in decreasing enterprises costs and offer an effective implementation of e-commerce activities, and solves the problem of enterprises due to lack of hardware and software assets. Study [7] highlighted the role of cloud computing in implementing successful e-commerce activities by SMEs.

Table 1. Previous studies

Article	Sample/methodology	Main results
Sultan, N. A. (2011)	British SME	flexible cost structure scalability.
Lin, A., & Chen, N.-C. (2012)	Interview of 19 Taiwanese IT professionals	compatibility with the companies' policy IS development environment Needs.
Brender, N., & Markov, I. (2013)	Sample of Swiss companies	company's size: larger companies is better fitting for the cloud.
Lian, J., Yen, D., & Wang, Y. (2014).	Survey of 106 CIOs of hospitals ANOVA	CIO innovativeness data security compatibility top manager support adequate resource perceived industry pressure
Alshamaila, Y. Papagiannidis, S. Stamati, T.(2013)	104 Greece organization online survey	only relative advantage
Gupta, P., Seetharaman, A., & Raj, J. (2013)	Survey of 211 SMEs/SMBs	ease of use convenience security and privacy cost reduction. Reliability sharing and collaborating.
Alkhater ,N. Wills ,G. Walters, G (2014)	twenty IT experts in Saudi Arabia semi-structured interviews.	Security privacy trust compliance
Harfoushi, et al (2016)	223 questionnaires were distributed to the IT departments of different Jordan hospitals	Technological factor have the most impact on the adoption decision.
Makena, J (2013)	220 interviews with ICT officers and business owners In Kenya	technological, organizational and environmental factors

Aydin in [11] presented how cloud computing was started as a theoretical concept and later on was applied in many areas like e-commerce domain. In his study he discussed how Internet computing; another name for cloud computing; influenced e-commerce applications. Desai in [12] explained how mobile applications along with deploying of cloud computing provide great accessibility for e-commerce enterprises where majority of employees and customers can use hand held devices for various

business tasks which reflected in lots of advantages for businesses with lower costs and high benefits. Shah in [13] claimed that SME in the last two decades suffer from many challenges related to lack of proper resources, security, and high implementation costs. He discussed how cloud computing was used to solve such problems.

3. THEORETICAL BACKGROUND AND RESEARCH MODEL

There are a lot of computer fields where adoption theory is being applied to determine the factors that drive technology acceptance and implementation. For example, study [14] investigated the factors in bank industry that leads to adopt internet banking in comparison with various traditional delivery machines. Study [15] studied the influences of adopting e-learning education from the perspective of learners and instructors. In addition, study [16] applied UTAUT theory to investigate the practitioners' behaviour in adopting different forms of social media for public relations by non-profit organizations. In addition, study [17] applied the technology acceptance model to examine the adoption of online shopping across cultures. With regard to the evaluation of specific perceptions of technologies, a number of researchers have described the motivation behind the choice of recommending a new technology to some reasons like cost-effectiveness, security, reliability and organizational needs [22]. In addition to those factors, Roberts and James [18] added a non-technical factor: customer service.

Dachyar and Prasetya [19] studied the adoption of cloud computing in Indonesian business companies. Results of their study showed that security, need, cost and reliability were significant in decision to adopt cloud computing. These factors were also found to be significant in study [20]. Gupta [21] found that the use of cloud computing can improve the organization business security and reduced costs. Security control and threat detect techniques must be well managed by cloud providers to engage customer trust in cloud services in an ongoing process in involve high level security standards [23], since cloud security still a doubtful model it must be controlled through several perspectives: cloud architecture, stakeholders perspective like applying complex access requirements and service delivery models perspective [24].

High reliable cloud service is concerned with high operation and maintenance cost in adopting well defined backup strategies, hardware failure strategies, security hardware and technology used and system status monitoring techniques, so cloud provider have to make a trade off to gain customer expectations of reliability level obtained and remain attractive to the customer as a cost reduction service model [25].

Furthermore the virtualization feature used by cloud computing to improve hardware resources efficiency increase the reliability issue [26]. Cost minimizing while performance is guaranteed is attractive feature of cloud pay-per-use model, when the organization pay only for operational processes and data space cost [27] which allow the e-commerce to scale up while the business improve, or scale down or maybe close without the loss of a purchased infrastructure. The need of cloud computing is concerned with considering the cloud advantages and services which is still unclear for many organizations. Hence, some efforts from the cloud computing providers are required to clarify such benefits [28].

The adoption of could computing in Jordan is still in its early stages. There are a lot of businesses and government agencies are still reluctant toward deploying it for many reasons. Some of these reasons are related to the company itself whereas some other reasons are related to implementing the technology itself. Furthermore, building and developing IT infrastructure with high capacity need a well-skilled staffs in IT fields to procure and build hardware, software, networks, and databases which serve the

needs of the organizations and their needs to support activities that enhance and accelerate business processes and ultimately improve business performance as well as productivity [30].

The term IT infrastructure is a comprehensive term that composed of physical and virtual resources that an organization's IT infrastructure can support. This infrastructure is determined by many major components like: hardware, software, networks and communications facilities (including Internet and Intranet), and database and its users [30].

Cloud computing is considered as one of the IT capabilities that help businesses to be more agile and allow them to achieve the advantage of using the latest technologies of hardware and/or software delivered by a cloud vendors via the Internet. This service can be included within the strategy of business companies. Globally, many companies and large governmental institutions, including private companies in modern countries like the United States of America went to cloud computing and had a positive impact on these institutions in terms of performance, efficiency, productivity and lower operating costs. Turban et. al [30] explained some benefits that can be achieved by implementing cloud computing like improving assets utilization, agility, and shifting from asset ownership to service management. This enables companies to get rid of maintenance cost and concentrates on internal issues. In Jordan, many institutions and companies are concerned about various types of issues related to financial aspects, data privacy and confidentiality when the matter related to IT deployment. This may be one of the biggest obstacles that prevent such companies from migrating their data to external storage media or using an external host to accomplish their day-by-day daily tasks. Such tasks are evolved around activities like: supplier relationship process, new service/product development process, order fulfillment, and customer relationship [31].

In order to achieve the goals of this research, the authors have conducted many visits to many e-commerce SMEs in Jordan. During the interviews that were held with many managers of SMEs, the conversations were focused on many factors that are supposed to be considered as benefits and challenges of adopting of cloud computing. One example of the interviews was held with Mr. Majd Jama'ah, the infrastructure engineer at Opensooq; the largest e-commerce enterprise in Jordan; explained that the need of cloud is that you can scale up or down your all infrastructure easily. This can be done because the data center is in the cloud and consequently, this offer the mobility feature to e-commerce organization. This effects reduce the costs in any company adopt cloud computing, so companies can just pay for what the really use, no extra or idle resources. This reinforces the fact that enterprises should have IT architecture that make a good balance or alignment between needs and technology. Further, security measures that are adopted by companies that offer cloud computing may be more strong than what is really used in some SMEs. The reliability depends on that where your cloud is, there are variant companies with variant reliable service.

Understanding benefits of cloud computing determines the extent of adoption of cloud computing and identify the functions where it is applicable to improve predetermined performances of the functions. Based on studies [7]-[9], cloud computing supports cost saving, scalability, availability, innovation speed, efficiency, easy management and disaster management.

Based on the presented information, this study identify four main factors: security, cost, reliability, and need as a determinant

factors that affect cloud computing adoption process by SME managers in Jordan. Figure.1 illustrates the research model.

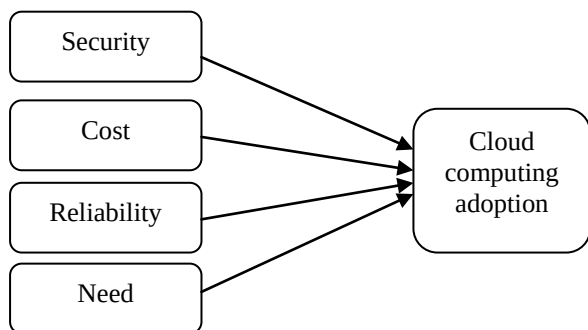


Figure 1. Research model.

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4. RESEARCH METHODOLOGY

Given the shortage of empirical studies in the area of cloud computing adoption in Jordan and the necessity to obtain concise data that describes the reality and nature of deploying of cloud computing in e-commerce SMEs, hence, this study is considered exploratory in nature. The process of data collection started on November 2017 till end of December 2017.

4.1 Instrumentation Description

The survey instrument used in this research was developed as open questions via interviews. The ideas of the open questions of the instrument are based on ideas from study [22]. Open questions were used to enable the interviewee to express all his/her ideas about the given questions. To improve the reliability of the instrument, two experts in the field of cloud computing have been consulted. Based on the given feedback, questions in their final forms were prepared for the planned interviews. Later on, a common interview protocol was prepared that enables the researchers to explain the purpose behind this study and setup a predefined date and time for the interviews.

The instrument of the study formulated of three parts: the first part consists of some general questions about the company: number of employees, establishment date, purpose of the company and, IT assets. The second part of questions pointed to the nature of using cloud computing and benefits that the company may gain from the deployment of this technology. The third part highlights the main challenges and difficulties that prevent or inhibit the company from adopting cloud computing. Sixteen SMEs were visited by the researchers for the benefit of this study.

4.2 Data Analysis and Research Results

After finishing the process of data collection, data are manually prepared and categorized based on sections that are presented during the interviews. Data analysis of the collected data revolved around two main baselines. The first one related to some internal issues of the company while the other baseline represents issues related to cloud computing service providers.

Some organizational factors like need and cost were found as major benefits of deploying cloud computing in e-commerce SMEs that were covered in the study. E-commerce SMEs are

willing to decrease the costs and to use latest technologies in terms of software and hardware according to their needs which align their IT architecture. This result assists the fact that companies can reduce their costs and utilize strong IT infrastructure that can be provided by cloud computing service provider. As a result, e-commerce SMEs can offer better services to its customers as well as enhancing its business model.

Security and reliability were found as a two-sided sword in this study. 71.4% of interviewed managers indicated that large companies who provide cloud computing services can offer strong security mechanisms to the cloud data. In addition, hosted data can be accessed remotely by companies' employees and business partners. In addition, deploying latest technologies in terms of strong and fast processors can increase companies processing powers and increase reliability. On the other side, 14.3% of managers are reluctant of using cloud computing as they are not confident about the security measures that will be used by the service provider; hence the company will be under a continuous risk. Accessibility will be considered also as another problem. In sum, this study indicated that the four factors: cost, security, reliability and need were found as major determinants for the adoption of the cloud computing in the visited enterprises.

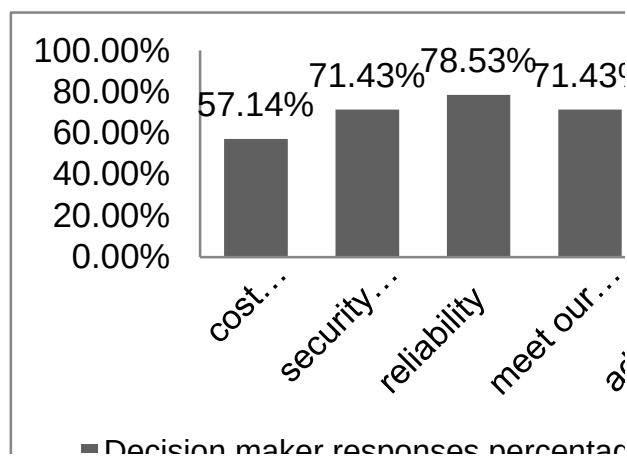


Figure 2. Respondent's ratio of factors related to cloud computing adoption.

Companies which are visited in this study agreed that cloud computing approach is an important force that can be adopted to have considerable cost saving over traditional computing methods. This benefit yields to have IT architecture according to the enterprise needs without any extra technologies or machines. Analysis of the data showed that 78.5% of managers agreed about reliability of using cloud computing in their organizations as employees are able to access the enterprise resources remotely. The effect of four factors is shown in Figure 2.

Consequently, mobility and availability are both achieved. Security was also an important factor in this domain as enterprises data should be secured and managers are aware of the advance techniques that can be deployed by many cloud computing providers like oracle, IBM, and Google as leading companies in this field.

Through the research open question interviews, a comparison between cloud computing and traditional technology was conducted, and Figure 3 addressed this comparison, and shows a highly trust in cloud reliability comparing with traditional technology by organization decision makers. In addition, the

response percentage for each factor for cloud technology was higher in a clear difference values.

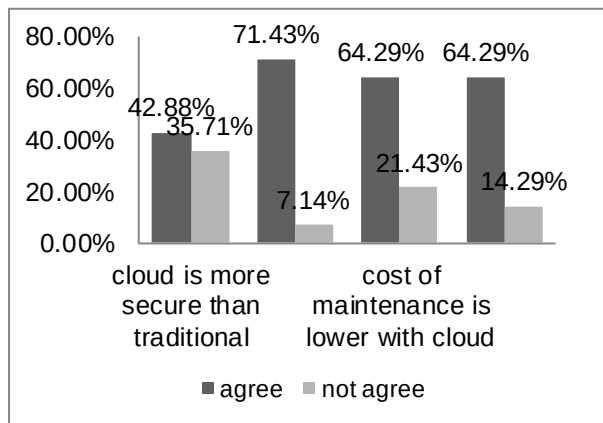


Figure 3. Cloud technology in comparison with traditional technology.

5. CONCLUSION

Cloud computing offers several benefits for e-commerce enterprises. But, the idea of migrating an existing system to the cloud is affected by organization's manager decision. There is a shortage of empirical studies in Jordan that focuses in adoption of cloud computing in e-commerce SMEs. As such, this study is one among the rare studies which were implemented in Jordan. The main purpose of this research was to investigate the deployment of cloud computing in e-commerce SMEs in Jordan.

The study has focused on two main silent domains: factors that are related to the company itself and considered as internal factors. The other part was concentrating on factors which are related to the service provider of cloud computing. From another point of view, this research explains how to encourage organizations to adopt cloud computing services by investigating the security and cost saving factors that influence an organization's intention to adopt cloud computing. Hence, from SMEs managers' point of view; this research was able to answer the motivations of deploying this new paradigm by e-commerce SMEs.

Results of this study showed that organizational needs and costs were found to be important and significant factors for deploying cloud computing technology in SMEs. These factors are considered as internal factors. On the other side, security and reliability also found to be a determinant factor for using this technology. For a future research, the researchers may consider main factors that are related to SME's environmental and technological domain. In addition, a more comprehensive plan can be put up to do more visits for more investigation.

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